

Erwin Rodrigues

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

C++ software engineer focused on electronic markets, with two front-office software engineering internships and applied AI project experience.

EDUCATION

EPITA <i>Computer Science, Major in Data Science / Artificial Intelligence</i>	Paris, France Sep. 2020 – Aug. 2026
INHA University <i>Exchange Program - Programming (C++, OOP, Systems) and Mathematics, GPA: 4.1</i>	South Korea Feb. 2022 – June 2022

EXPERIENCE

BNP Paribas CIB - Global Market Access <i>Software Engineering Intern - Front Office - FORCE Team</i>	March 2026 – Aug. 2026 Paris, France
- Extended production C++ session monitoring with connection and disconnection tracking and static client information across multiple protocols - Used C++20 Concepts and variadic templates to extend session metadata with compile-time type checks; added automated non-regression tests - Prototyped message-count and latency monitoring with isolated handlers and asynchronous event dispatch; validated locally but not deployed - Migrated a market interface to updated core libraries by upgrading dependencies and adapting code; built a market-specific simulator to check expected responses as an alternative to UAT	
Capital Fund Management (CFM) <i>Software Engineering Intern - Front Office</i>	Sep. 2024 – Jan. 2025 Paris, France
- Built a from-scratch C++ mock-exchange proof of concept for testing trading applications; the project did not reach production - Implemented an order book and matching engine supporting order changes, partial fills, and time-in-force behavior - Made broker and event mappings configurable to generate tailored FIX responses for scenario tests - Wrote GTest coverage for order management, matching behavior, and FIX message generation	

PROJECTS

Big Data – Data Pipeline and Dashboard <i>Python, Docker, TimescaleDB, Dash</i>	2025
- Built ETL pipelines to extract, clean, and reconcile multi-source equity data (Euronext, Boursorama) into TimescaleDB hypertables with batched/parallel processing - Developed an interactive Dash dashboard with log-scale charts, candlesticks, multi-asset comparison, Bollinger Bands, and volume analysis across markets (Paris, NYSE, LSE) - Containerized the stack with Docker for reproducible, modular deployment (ETL, database, and dashboard)	

COMPETITIONS

Institut du Cancer de Montpellier - PinkCC Challenge <i>Deep Learning</i>	April 2025 - June 2025
- Benchmarked ovarian-cancer CT segmentation models; continued ovseg training and ran Molmo training and QLoRA fine-tuning experiments on A100 GPUs - Team placed 8th in the challenge's technical ranking	
Banque de France Hackathon <i>TypeScript, React, AI Agents</i>	June 2025
Designed and developed an application using React that builds newsletters leveraging AI Agents	